

AI Management Consultant

Project documentation

1. Introduction

The AI Management Consultant is a multi-agent artificial intelligence system developed to support businesses in analyzing their data and making informed strategic decisions. The system provides an interactive platform where users can upload business data in CSV or Excel format and describe their business problem or objective. It is designed to convert raw business data into meaningful information that can be easily understood by management.

The system performs different types of business analysis, including revenue, cost, units sold, inventory, marketing spend, and product performance analysis. It also provides data-quality checks, trend analysis, segmentation, and analytical metrics. These features help identify important business patterns, unusual changes, and areas that require attention.

The project uses a multi-agent AI approach, where specialized agents perform different tasks such as understanding requirements, planning the analysis, processing and analyzing data, testing results, reviewing insights, and preparing documentation. The agents work together through an organized workflow to produce more structured and useful business analysis.

Based on the analysis, the system generates AI insights, risk assessments, strategic recommendations, action plans, and executive reports. The final results can also be generated as a downloadable PDF report. Thus, the AI Management Consultant provides a complete solution for transforming business data into actionable insights and decision-support information.

2. Problem Analysis

2.1 Business Context

Modern businesses generate large amounts of data from sales, costs, inventory, marketing activities, and product performance. Analysing this information manually can be time-consuming and makes it difficult for management to identify important trends, risks, and opportunities.

Business teams need to evaluate different factors such as revenue, units sold, product profitability, inventory levels, and marketing spend before making strategic decisions. However, these activities may involve multiple analysis steps and require significant manual effort.

Some of the major challenges include:

- Difficulty analysing large amounts of business data
- Time-consuming manual analysis
- Difficulty identifying unusual trends and performance changes
- Lack of centralized business insights
- Difficulty assessing business risks

- Manual preparation of recommendations and reports

The proposed AI Management Consultant addresses these challenges by using a multi-agent AI approach. Specialized agents collaborate to process business data, perform analysis, validate results, generate insights, and prepare strategic recommendations and reports.

3. Stakeholders

3.1 Business Management

Business managers are the primary decision-makers who use the system to:

- Monitor overall business performance
- Review revenue, cost, and profitability
- Analyse inventory and marketing spend
- Identify business risks and opportunities
- Review strategic recommendations
- Support data-driven decision-making

3.2 Business Analysts

Business analysts use the system to:

- Upload CSV/Excel business data
- Define business problems and objectives
- Analyse business metrics and trends
- Identify product performance
- Review AI-generated insights
- Generate business reports

3.3 System Users

System users interact with the application to:

- Upload business datasets
- Enter business queries
- View dashboards and analysis
- Check data quality
- View insights and recommendations
- Download generated reports

3.4 AI Agents

AI agents perform specialized tasks such as:

- Understanding business requirements
- Planning the analysis workflow
- Processing and analysing data

- Testing and validating results
- Reviewing generated insights
- Preparing documentation and reports

3.5 Project Developers

Project developers are responsible for:

- Developing the application
- Implementing the multi-agent workflow
- Managing data-processing components
- Testing the system
- Maintaining and improving the application

4. Existing Problems and Challenges

The existing business analysis process often depends on manual data handling and separate analytical activities. This can make it difficult to obtain accurate and timely insights for decision-making.

Major Challenges

- **Manual Data Analysis** – Requires significant time and effort.
- **Data Quality Issues** – Missing or duplicate data can affect analysis.
- **Limited Profitability Visibility** – Difficult to identify the most and least profitable products.
- **Scattered Business Information** – Sales, costs, inventory, and marketing data may need to
- **Trend Identification** – Unusual changes in revenue, costs, or product performance can be difficult to detect manually.
- **Risk Identification** – Business and operational risks may not be identified quickly.
- **Report Preparation** – Creating detailed business reports and recommendations manually is time-consuming.
- **Decision-Making Delays** – Lack of consolidated insights can slow down strategic decision.

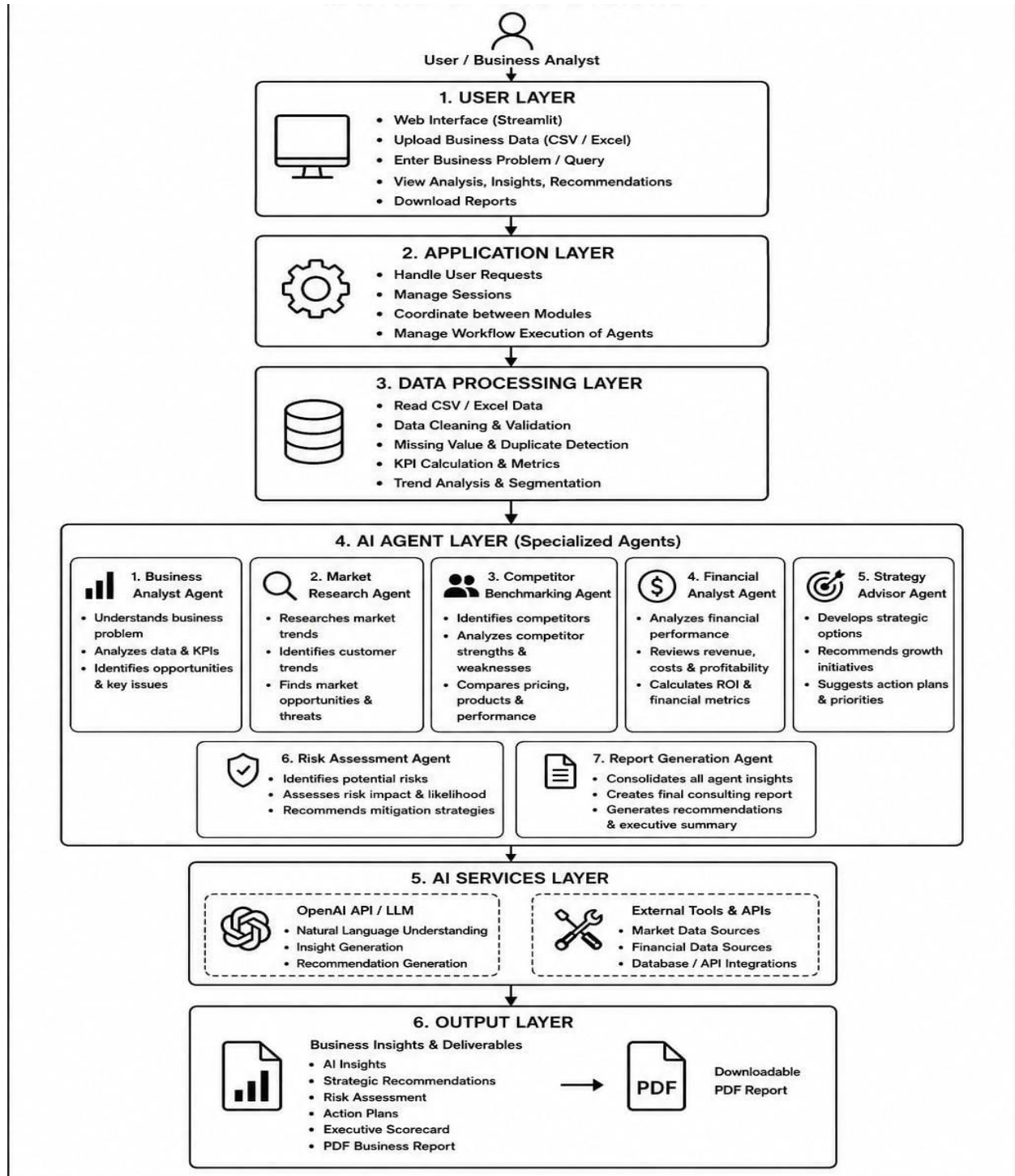
5. Project Objectives

The main objectives of the AI Management Consultant project are:

- Reduce manual effort involved in analysing business data.
- Analyse CSV/Excel data related to revenue, cost, sales, inventory, and marketing.
- Provide important metrics such as revenue, units sold, cost, marketing spend, inventory, and product performance.
- Assign specialized tasks to different AI agents for structured analysis.
- Detect trends, performance patterns, and areas requiring attention.

- Identify potential financial, market, operational, and customer-related risks.
- Provide strategic recommendations and practical action plans.
- Generate structured executive reports in PDF format.
- Help management make faster and more data-driven business decisions.

6. Architecture Diagram



The Architecture diagram shows how the AI Management Consultant processes a business problem from user input to the final consulting report. The system is divided into different layers, where each layer performs a specific function.

6.1 User Layer:

The user interacts with the system through the Streamlit web interface. The user can upload business data in CSV/Excel format, enter a business problem or query, view analysis and recommendations, and download the generated report.

6.2 Application Layer:

This layer handles user requests and manages the overall application workflow. It coordinates the different modules and controls the execution of the AI agents according to the business problem.

6.3 Data Processing Layer:

The uploaded business data is read and processed in this layer. It performs data cleaning and validation, identifies missing or duplicate data, and calculates important KPIs and business metrics required for analysis.

6.4 AI Agent Layer:

This is the core of the system. It contains seven specialized agents:

Business Analyst Agent – understands the business problem and analyzes KPIs.

Market Research Agent – analyzes market trends, opportunities, and threats.

Competitor Benchmarking Agent – studies competitors, pricing, strengths, and weaknesses.

Financial Analyst Agent – analyzes revenue, costs, profitability, and ROI.

Strategy Advisor Agent – develops strategic recommendations and action plans.

Risk Assessment Agent – identifies risks and suggests mitigation strategies.

Report Generation Agent – combines the results and prepares the final consulting report.

6.5 AI Services Layer:

The agents use the OpenAI API/LLM for language understanding, analysis, insight generation, and recommendations. External tools and APIs can also provide additional market or financial data.

6.6 Output Layer:

Finally, the system converts the analysis into useful business deliverables such as AI insights, strategic recommendations, risk assessment, action plans, executive scorecards, and a downloadable PDF business report.

Overall flow:

User Input → Application → Data Processing → Specialized AI Agents → AI Services → Business Insights & Recommendations → Final PDF Report.

7. Multi-Agent Design

7.1 Agent Architecture Introduction

The AI Management Consultant uses a multi-agent architecture in which different AI agents are assigned specialized business analysis tasks. A Consulting Manager Agent coordinates the overall workflow and delegates the user's business problem to the appropriate specialist agents.

Each specialist agent performs a specific task and produces structured results that can be used by the next stage of the analysis. This approach separates responsibilities between agents and helps generate a comprehensive business consulting response.

The architecture consists of Business Analyst, Market Research Agent, Benchmarking Agent, Financial Analysis Agent, Strategy Advisor, and Report Writer, coordinated by the Consulting Manager.

7.2 Role of Each Agent

7.2.1 Business Analyst Agent

Understand the user's business problem and identify the key information required for analysis.

Responsibility:

- Understand the business problem
- Identify objectives and key issues
- Analyze business data and KPIs
- Identify opportunities and possible causes

Output:

Business problem summary, objectives, key KPIs, and business insights.

7.2.2 Market Research Agent

Analyze the market environment related to the given business problem.

Responsibility:

- Research current market trends
- Analyze customer trends
- Identify market opportunities
- Identify market threats
- Study relevant industry factors and competitors

Output:

Market insights, trends, opportunities, and threats.

7.2.3 Benchmarking Agent

Evaluate the competitive position of the business.

Responsibility:

- Identify major competitors
- Analyze competitor strengths and weaknesses
- Compare pricing and products
- Identify competitive opportunities

Output:

Competitor analysis, benchmarking results, and competitive gaps.

7.2.4 Financial Analysis Agent

Evaluate the financial aspects of the business problem.

Responsibility:

- Analyze revenue-related issues
- Analyze major cost factors
- Evaluate profitability
- Calculate percentage changes
- Calculate ROI and financial metrics

Output:

Financial insights, profitability analysis, ROI, and financial metrics.

7.2.5 Strategy Advisor

Convert the analysis into practical business strategies.

Responsibility:

- Identify strategic priorities
- Develop recommended actions
- Suggest short-term and long-term strategies
- Estimate expected business impact
- Identify relevant KPIs

Output:

Strategic recommendations, priorities, and action plans.

7.2.6 Report Writer

Combine the outputs from the different specialist agents into a final consulting response.

Responsibility:

- Consolidate agent results
- Organize key findings
- Summarize important insights
- Prepare the final consulting report

Output:

Final consulting report and executive-level summary.

7.2.7 Consulting Manager Agent

Coordinate the complete multi-agent consulting workflow.

Responsibility:

- Understand the user's request
- Select the appropriate specialist agents
- Delegate tasks using agent handoffs
- Coordinate the results
- Ensure the final response is useful and consistent

Output:

Coordinated consulting workflow and consolidated final response.

7.3 Agent Interaction and Handoff Flow

The workflow begins when the Consulting Manager Agent receives the user's business problem and determines which specialist agents are required.

The typical flow is:

User Problem → Consulting Manager → Business Analyst → Market Research → Benchmarking → Financial Analysis → Strategy Advisor → Report Writer → Final Consulting Report

Each agent performs its specialized task and passes its result to the next relevant agent. The Consulting Manager acts as the central coordinator and uses handoffs to delegate work between agents.

This allows the system to break a complex business problem into smaller specialized tasks and then combine the results into a single consulting response.



7.4 Tools Integration Overview

The project integrates tools with specific agents to improve their analytical capabilities.

Tool/ technology	Used for	Agent
WebSearchTool	Market and industry research	Market Research Agent
Calculate_percenatge_change	Percentage/ financial change calculations	Financial Analysis Agent
Calculate_roi	ROI calculations	Financial Analysis Agent
Pandas	Business data processing and analysis	Data Analysis
OpenAI Agents SDK	Agent creation, coordination and handoffs	All agents
ReportLab	PDF report generation	Report Writer

8. Implementation

8.1 User Interface

The system provides an interactive user interface through Streamlit. Users can upload business data in CSV/Excel format and enter their business problem or query. The interface displays the generated analysis, insights, recommendations, and reports in an easy-to-understand format.

8.2 AI Agent execution

Each AI agent performs a specialized task.

The execution order of agents is:



8.3 Management and Context Management

The system maintains relevant business information and agent results throughout the consulting workflow.

Its responsibilities include:

- Business problem context
- Agent result sharing
- Data consistency
- Information handoff
- Maintaining relevant analysis context

The context is passed between the Consulting Manager and specialized agents so that each agent can build on the results produced by the previous stage.

8.4 Structured Outputs

The system organizes the responses generated by different agents into clear and consistent sections.

The structured results include:

- Business analysis
- Market research findings
- Competitor benchmarking
- Financial analysis
- Strategic recommendations
- Final consulting report

The structured results make it easier to combine outputs from multiple agents and present the final business analysis in a readable format.

8.5 Human Approval

The system allows the user to review the generated business insights and recommendations before using them for decision-making.

Human review includes:

- Reviewing generated insights
- Checking recommendations
- Validating important findings
- Confirming the final business report
- Making the final decision based on the AI output

9. Advanced Features

9.1 Multi-Agent Collaboration

- Uses multiple specialized AI agents.
- Assigns different business tasks to different agents.
- Enables collaboration through agent handoffs.
- Combines specialized analysis into one consulting response.

9.2 Agent Handoff Mechanism

- Consulting Manager coordinates the workflow.
- Delegates tasks to appropriate specialist agents.
- Transfers relevant information between agents.
- Reduces the need for manual task coordination.

9.3 Web-Based Market Research

- Uses web search for market research.
- Identifies current market trends and opportunities.
- Supports industry and competitor analysis.
- Provides additional information for business decision-making.

9.4 Financial Analysis Tools

- Calculates percentage changes.
- Calculates Return on Investment (ROI).
- Supports revenue and cost analysis.
- Helps evaluate business financial performance.

9.5 Automated Report Generation

- Consolidates results from multiple agents.
- Generates structured consulting findings.
- Supports PDF report generation.
- Provides executive-ready business insights.

10. Future Enhancements

10.1 Real-Time Data Integration

- Connect real-time market data sources.
- Integrate live business data through APIs.
- Enable continuously updated analysis.

10.2 Advanced Forecasting

- Add sales forecasting.
- Predict future revenue and demand.
- Identify potential business trends.

10.3 Database Integration

- Connect the system with databases.
- Store business data and previous analysis.
- Enable retrieval of historical business information.

10.4 Advanced Competitor Intelligence

- Add live competitor data.
- Track competitor pricing and product changes.

- Provide more detailed competitive insights.

10.5 More Specialized Agents

- Add dedicated risk assessment agents.
- Add forecasting agents.
- Add customer analytics agents.
- Add supply-chain or inventory analysis agents.

10.6 Automated Recurring Reports

- Generate reports automatically at scheduled intervals.
- Track changes in business performance.
- Provide regular executive summaries.